

Pay-Equity Audit with Supervised Learning

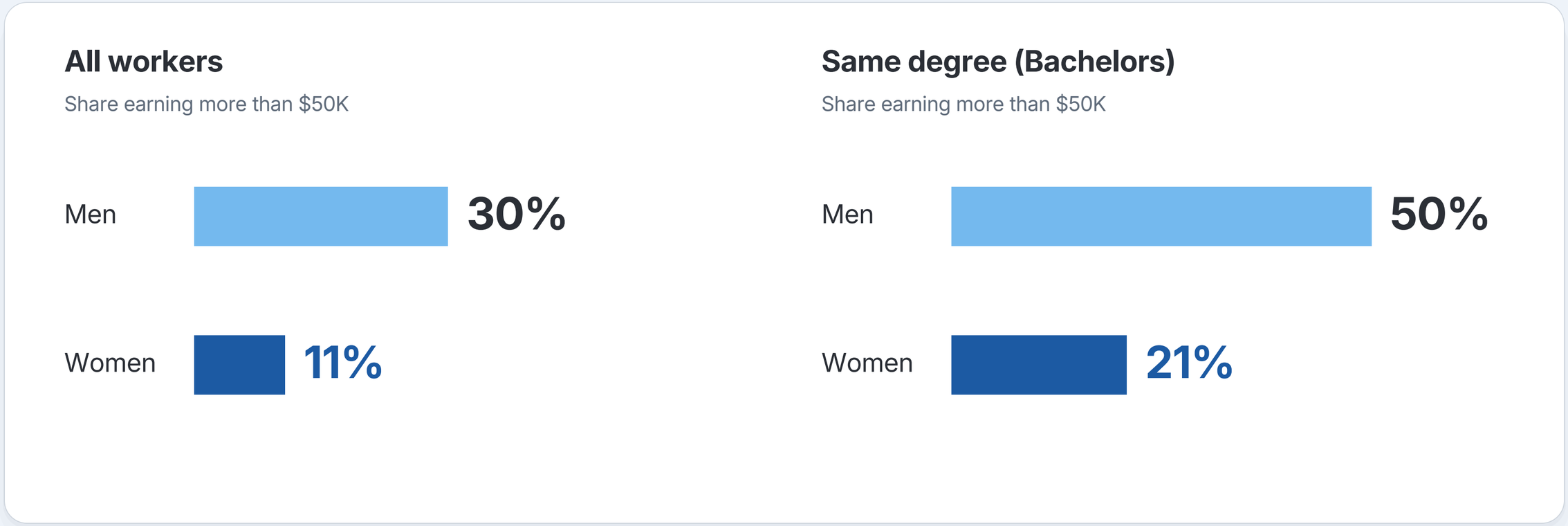
Our role: a pay-equity regulator

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Supervised Learning & Model Engineering · Ilana Sebag · Albert School · October 2026

Is the pay gap explained by the job?

Our questions: Which employers should we investigate? Does the gap remain for equal work? How sure can we be?



Equal pay for equal work → compare each person with what their job-related profile predicts.

Our data and what the model may use

USED · job-related factors

- Age
- Education (years)
- Occupation
- Hours per week
- Workclass
- Investment income

EXCLUDED · protected attributes and proxies

- ~~Sex~~
- ~~Race~~
- ~~Native country~~
- ~~Relationship~~
- ~~Marital status~~

> Husband = 99.99% men:
> relationship encodes sex

+5 features we created, e.g. part-time work: 28% of women vs 12% of men, a legitimate reason to earn less.

Adult census (1994) · 48,842 people · only 24% earn > $50K → we judge models on average precision, not accuracy

One pipeline, three models

Raw data → Feature engineering +5 new features → Imputation + one-hot → **Model**

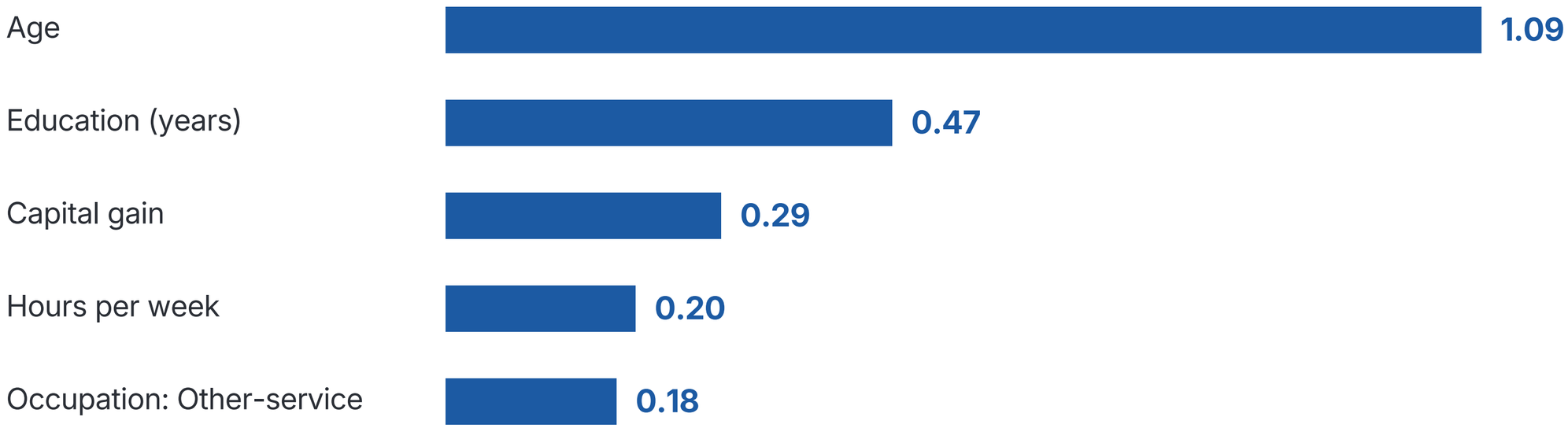
Average precision on the validation set (main metric: focuses on the rare >$50K class)

Logistic Regression	Random Forest	LightGBM
0.692	0.752	0.776
		Selected: tuned LightGBM

Optuna + 5-fold cross-validation on the training set · every trial logged in MLflow · test set used only once

What drives the predictions (SHAP)

Mean |SHAP value|: average impact on the prediction (log-odds, 2,000 validation people)

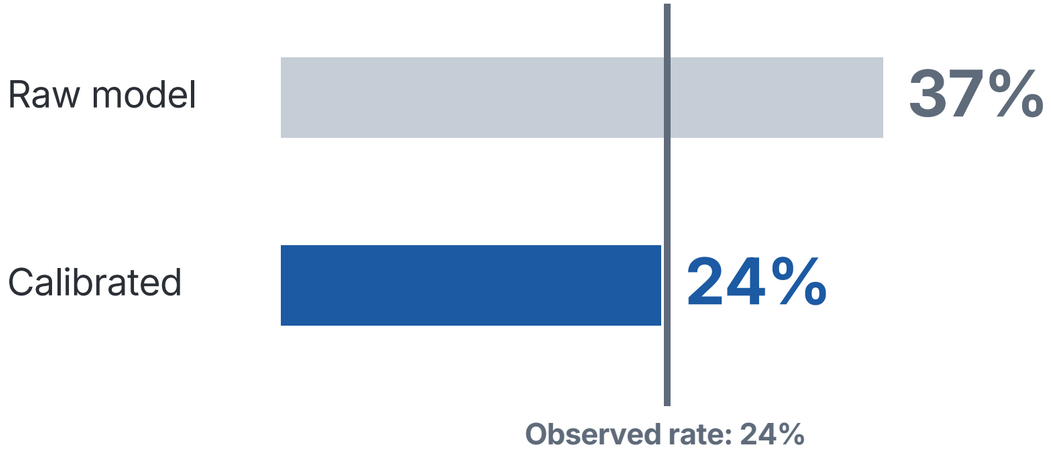


→ **Only legitimate pay factors: experience, qualification, investment income, working time, job.**

SHAP describes what the model uses, not what causes income.

Can we trust the probabilities?

Mean predicted probability vs observed rate



Calibration error (ECE)

~~0.135~~

0.009

raw model on validation → isotonic, untouched test set (validation: 0.010). Class weights made the raw model over-confident.

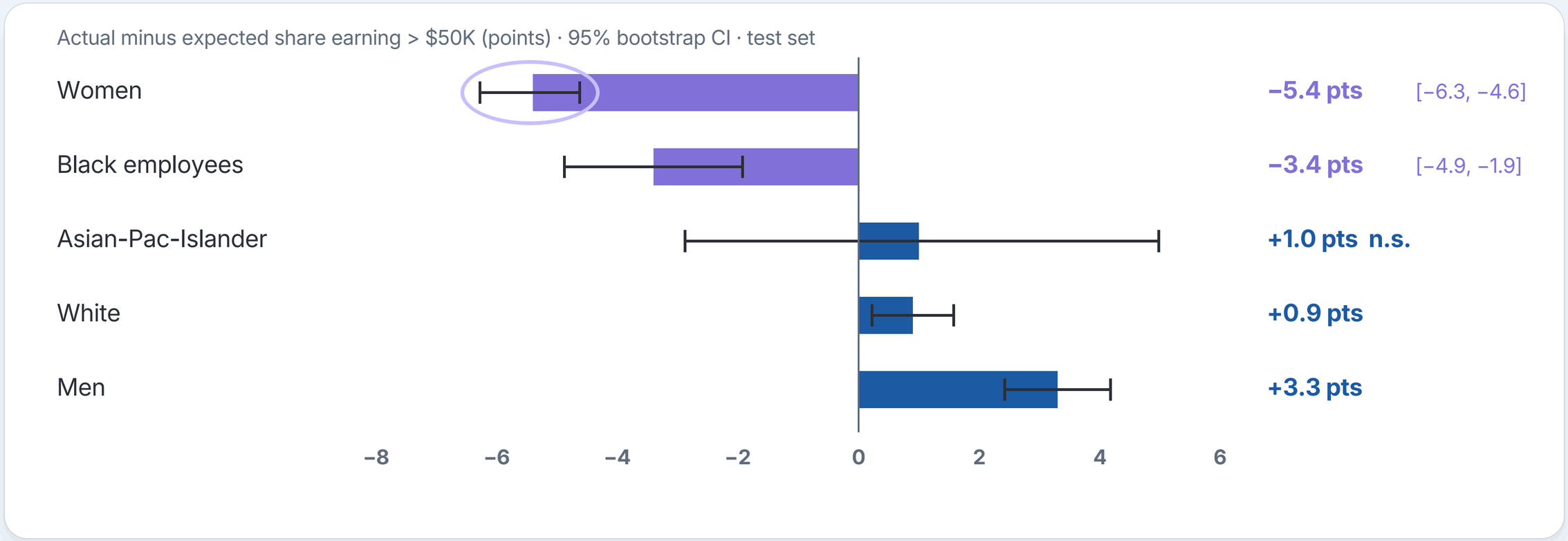
Calibrated probabilities = a fair expected rate per group.

Isotonic calibration learned on out-of-fold training predictions: validation and test sets untouched.

The unexplained gap

−5.4 pts

Women vs. what their job-related profile predicts



Unexplained ≠ proven: a reason to investigate, not a proof of discrimination.

Why not use all columns?

Before: all columns

incl. sex, race, relationship, marital status

ROC-AUC

0.930

Unexplained gap, women

+0.2 pts

After: job-related only (ours)

no sex, no race, no proxies

ROC-AUC

0.893

Unexplained gap, women

−5.4 pts

The full model is more accurate because it learned the discrimination, and it hides it.

Same tuned LightGBM, calibration and threshold; only the input columns differ (test set).

From model to tool

FastAPI service

POST /audit

An inspector sends a company's employee records → gap per group, 95% CI, flagged employees

```
{ "women": { "gap": -6.0, "ci": [-7.8, -4.0] },
  "flagged_underpaid": 52 }
```

Simulated company of 2,000 employees

POST /predict

One profile → expected probability. Same profile, any sex or race: same answer.

Limits

- **Unexplained ≠ proven**
- **1994 data**
- **Total income, not salary**

Use it to prioritise investigations.

Thank you!

github.com/rcacarie-png/Adult-Income-Prediction



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Thank you for listening

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